

$$\frac{bc}{be} = \frac{BC}{BE}$$

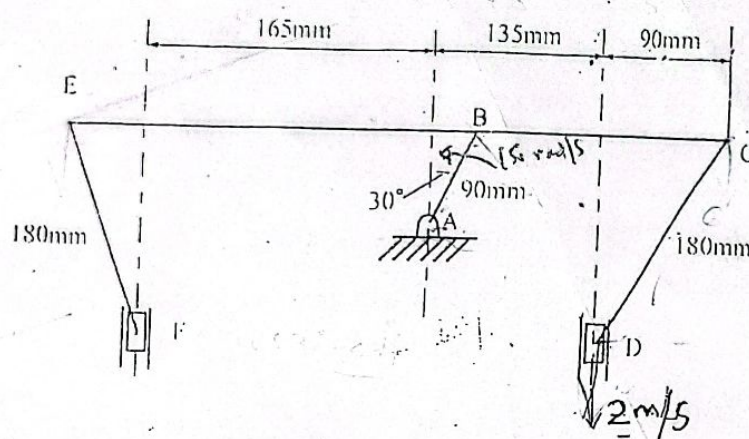
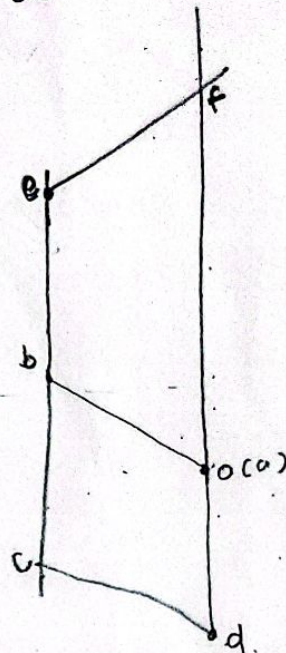


Fig. Q 4



CAT 1 a)

Explain what you understand by the following terms as used in planar mechanisms.

- (i) Kinematic pair
- (ii) Kinematic chain
- (iii) Degree of freedom
- (iv) Mechanism

(4 Marks)

b)

In the mechanism shown in Fig. Q 4, the crank AB revolves anticlockwise at 16 rad/s and the point D is constrained to move vertically downwards at 2 m/s. The link CE = 450 mm and is horizontal in the position shown. By drawing the velocity diagram or otherwise, determine:

- (i) The velocity of point B
- (ii) The velocity of point C relative to B
- (iii) The velocity of B relative to E
- (iv) The velocity of E relative to F
- (v) The rubbing velocity at E if the pin diameter is 30 mm
- (vi) The velocity of point F.

(16 Marks)

c)

Explain what you understand by the following terms as used in mechanisms.

- i) Mechanical advantage
- ii) Transmission angle
- iii) Instantaneous centre
- iv) Kinematic chain

(4 Marks)

v_{ba} is $\perp AB$, draw a line AB through a ;

For v_{bg} , draw a line through g parallel to the line of stroke of the slider B on the guide G .

The intersection of the two lines locates the point b .

Velocity of $B = gb = 2.67 \text{ m/s}$.

Example 2.7 In a mechanism shown in Fig. 2.15(a), the angular velocity of the crank OA is 15 rad/s and the slider at E is constrained to move at 2.5 m/s . The motion of both the sliders is vertical and the link BC is horizontal in the position shown. Determine the

- rubbing velocity at B if the pin diameter is 15 mm
- velocity of slider D .

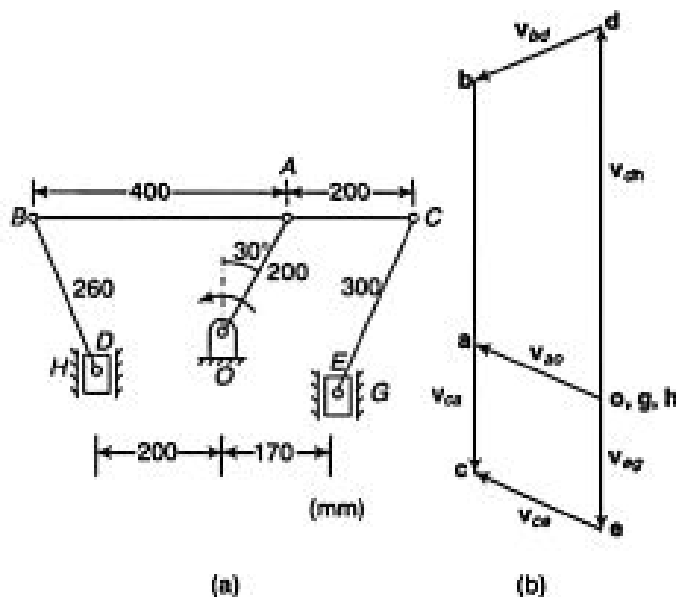


Fig. 2.15

Solution $v_a = \omega_a OA = 15 \times 0.2 = 3 \text{ m/s}$

Draw the velocity diagram as follows:

- Take vector oa to a suitable scale (2.15b).
- Consider two points G and H on the guides of sliders E and F respectively. In the velocity diagram, the points g and h coincide with o . Through g , take a vector g parallel to direction of motion of the

slider E and equal to 2.5 m/s using some scale.

- Through e draw a line $\perp EC$ and through a , a line $\perp AC$, the intersection of these two lines locates the point c .
- Locate the point b on the vector ca so that $ca/cb = CA/CB$.
- Through b , draw a line $\perp BD$ and through h , a line parallel to direction of motion of the slider D , the intersection of these two lines locates the point d .

- Angular velocity of link $BD = bd/BD = 2.95/0.26 = 11.3 \text{ rad/s}$ (counter-clockwise)
Angular velocity of link $BC = bc/BC = 8.4/0.6 = 14 \text{ rad/s}$ (clockwise)

Thus velocity of rubbing at

$$\begin{aligned} B &= (\omega_{bd} + \omega_{bc})r_b \\ &= (11.3 + 14) \times 0.015 \\ &= 0.38 \text{ m/s} \end{aligned}$$

- The velocity of the slider $D = hd = 8.3 \text{ m/s}$

Example 2.8 The lengths of various links of a mechanism shown in Fig. 2.16(a) are as follows:

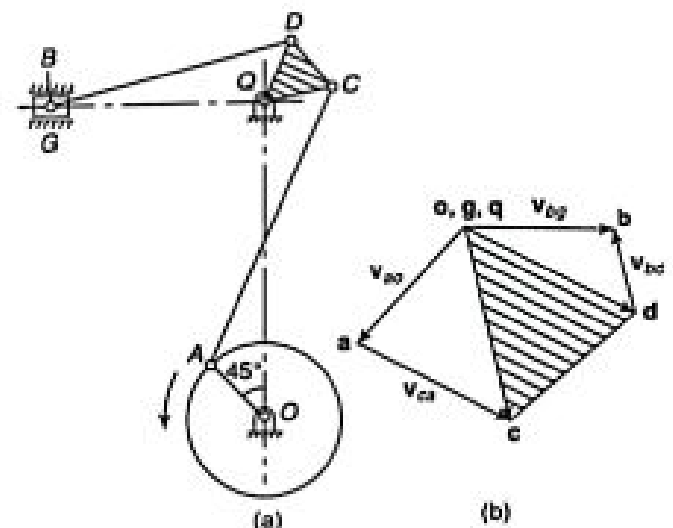
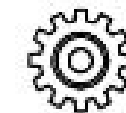


Fig. 2.16

$OA = 150 \text{ mm}$

$AC = 600 \text{ mm}$

$BC = 400 \text{ mm}$

$CD = 125 \text{ mm}$

$BD = 500 \text{ mm}$

$DE = 100 \text{ mm}$